



Instructor: Date:

Learner: Regional Branch:

Program: Center:

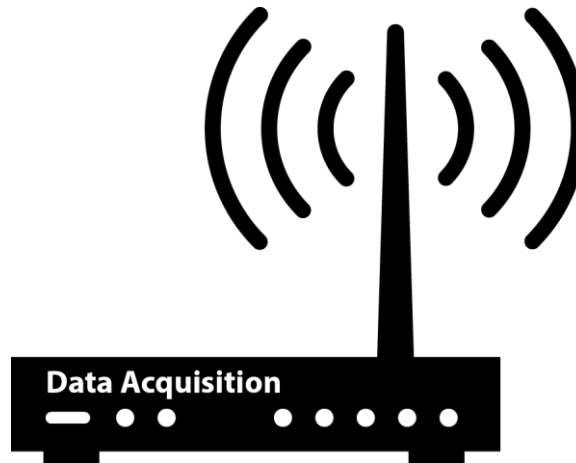
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Let's Start!

Data Acquisition Process



Learning Outcome:

Describe common activities at work to a visitor using short sentences in familiar spoken and written situations.
Understand information about data and its structure in simple spoken and written situations.

Ask your partner:

- How do you think a microphone works?
- What scientific instruments that collect data from the environment do you know?

Can Do Statements:

- I can explain what an acquisition system is.
- I can talk about the components of acquisition systems.
- I can give information about concepts related to data and software programming.





Let's Explore!

Ashley needs help to understand data acquisition systems.



Instructions: Read the text below.

Timmy: Hi, Ashley! What are you doing?

Ashley: I'm studying for my test, but it's too much information. You study engineering, right?

Timmy: Yes, I do. Why are you asking me that?

Ashley: I need your help. Do you know what an acquisition system is?

Timmy: Of course I do. A data acquisition system is what gathers real-time data and converts it into electrical signals thanks to a transducer, which has certain devices to produce an output that goes to a signal conditioning circuit. Then, a data acquisition device turns the signal into a stream of digital data.

Ashley: That doesn't sound that difficult. What are the components of a data acquisition system?

Timmy: Well, a data acquisition system may have many components. The first one, as I mentioned before, is a transducer which is a device that measures data and turns it into an equivalent electrical signal. The second component is a signal conditioning circuit which ensures that the signal fits into the range of the data acquisition device; it also isolates the signal from other waves coming from the computer, making the signal as clear as possible. Other important components are measurement hardware and control software. They are key elements for data acquisition.

Ashley: You are very smart, Timmy. Help me with one last thing, what is the most important component of data acquisition systems?

Timmy: I would say it is the sampling rate. This component is the number of times per second the device measures a signal. The more times it measures a signal, the clearer the representation. For example, sound, if the device measures a signal many times, the sound is better. We sometimes need much resolution to display data clearly.

Ashley: Thanks a lot, Timmy. I'm sure that after your explanation, I'll get a good grade.





Let's Practice!

Explaining What an Acquisition System Is

Instructions: Match column A with column B.

COLUMN A

1. What is a data acquisition system?
2. W How many components does a data acquisition system have?
3. What is the most important component of a data acquisition system?

COLUMN B

- a. A data acquisition system may have many components.
- b. The most important component is the sampling rate.
- c. It is what gathers real-time data and converts it into electrical signals thanks to a transducer.



Let's Practice!

Let's talk about components.

Instructions: Circle the correct word to complete the paragraph.

There are (many / much / one) components involved in data acquisition. We can divide these components into sensors, signal conditioning circuitry, and analog-to-digital converters. There is too (many / much / few) information about each component, but we can say that the whole data acquisition process is the result of converting and analysis of (many / much / a lot) electrical signals.





Extension Activity

Acquisition Systems

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Search for more information about acquisition systems.
- Identify common devices used for data acquisition.
- Make a list of 5 devices and write down their characteristics.
- Share your work with the class.

Self-Assessment

Yes

No

- I search for more information about acquisition systems.
- I identify common devices used for data acquisition.
- I make a list of 5 devices with their characteristics.
- I share my work with the class.

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Target Words

- **fit** (verb): to adapt
e.g. The second component is a signal conditioning circuit which ensures that the signal fits into the range of the data acquisition device.
- **ensure** (verb): to guarantee
e.g. The second component is a signal conditioning circuit which ensures that the signal fits into the range of the data acquisition device.
- **measure** (verb): to estimate the size or weight of something
e.g. The first one, as I mentioned before, is a transducer which is a device that measures data and turns it into an equivalent electrical signal.

- **engineering** (noun): study of pure sciences
e.g. You study engineering, right?

- **gather** (verb): to collect something
e.g. A data acquisition system is what gathers real-time data and converts it into electrical signals thanks to a transducer.



Learning Points

Countable and Uncountable Nouns with Many and Much

We use **much** and **many** to refer to a considerable quantity of a noun.

There are some nouns in software acquisition systems that we can quantify with **much** and **many** to express a great amount of something.

We use **much** to talk about a huge amount of an uncountable noun.

Example:

Much information

Much speed

We use **many** to talk about a big number of countable nouns.

Examples:

Many datum

Many signals

Let's take a look at some other examples of **countable** and **uncountable** nouns that we can use with **many** and **much**:

Countable Nouns	Uncountable Nouns
Many devices	Much information
Many transducers	Much resolution
Many instruments	Much ratio

Examples

- A data acquisition system may have many components.
- If the device measures a signal many times, the sound is better.
- We sometimes need much resolution to display data clearly.





Learning Points

Simple Present Tense and Present Continuous in Questions

We can ask questions in **simple preset** by using the auxiliary verb **do** or **does**. On the other hand, we use the verb **to be** as an auxiliary to create questions in **present continuous**.

We usually make questions in English by setting the **auxiliary verb** before the subject.

Examples:

Do you need a new system?

Are you sorting the requirements now?

The auxiliary verb of the simple present is **do** which becomes **does** in third person (*he, she, and it*).

Examples:

Does he know about data acquisition?

Do computers interpret data?

In **present continuous**, the auxiliary verb is the verb **be**.

Examples:

Is the software displaying the data?

Are we acquiring any data?

In the examples above, the first word is the auxiliary verb; when that verb is at the beginning of the question, we are looking for a **yes/no answer**, but we can also use **WH questions** to ask for information.

Examples:

Where do you work?

What type of data are you collecting?

Examples

- *What are you doing?*
- *Why are you asking me that?*
- *Do you know what an acquisition system is?*

